

DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY
SMA 3121 ::: COMPLEX ANALYSIS CAT 2 1 HOUR

- a) Given the triangle $A(0,1)$, $B(1,1)$ and $C(1,0)$. Using the transformation $T(z)=z^2$, discuss the Conformality of T at $C(1,0)$.
- b) Evaluate $\int_{1+i}^{2+4i} z^2 dz$, Along the straight line joining $(1,1)$ to $(2,1)$ then from $(2,1)$ to $(2,4)$
- c) Using contour integration and residues, evaluate $\int_0^{2\pi} \frac{\cos 3\theta}{5-4\cos \theta} d\theta$.
- d) From Schwarz-Christofel, show that $(\pi-\alpha_1)+(\pi-\alpha_2)+\dots+(\pi-\alpha_n)=-2$.
- e) Evaluate $\ell^{-1}\left[\frac{1}{(s^2+1)^2}\right]$ using the residue theorem.
- f) using Inversion mapping $f(z)=\frac{1}{z}$, transform the line $y=2x+3$

$$\begin{aligned} (s-i)(s+i) &= s^2 - i^2 \\ &= s^2 + 1 \\ &= (s-i)(s+i) \end{aligned}$$

$$\begin{aligned} x(x+iy) &= x^2 - y^2 + iyx \\ iy(x+iy) &= -xy^2 + x^2i \end{aligned}$$